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ABSTRACT

The increasing demand for intelligent and cost-effective security systems has encouraged the development of Internet of Things (IoT)-based monitoring solutions capable of providing real-time intrusion detection and remote supervision. Conventional security systems often require substantial financial investment and may lack flexibility, scalability, or continuous online monitoring capabilities. Consequently, there is a growing need for accessible security solutions that integrate embedded systems, wireless communication, and cloud technologies.

Project VANTA is an IoT-enabled laser tripwire security alarm system designed to detect unauthorised access within a protected area through the interruption of an optical boundary. The system is centred upon the ESP32 microcontroller, which continuously monitors a laser receiver sensor and processes intrusion events in real time. Upon detection of a breach, the system activates both audible and visual alarm mechanisms, updates a local display interface, and transmits status information to a cloud-connected monitoring dashboard.

The developed system incorporates a laser emitter module, laser receiver sensor, active piezoelectric buzzer, LED indicators, and a 16×2 I²C liquid crystal display. Wireless communication is achieved through the ESP32's integrated Wi-Fi functionality, whilst cloud-based monitoring is facilitated through the Blynk platform and a custom-developed web dashboard. An administrative bypass mechanism has also been implemented to permit authorised temporary disarming of the system for maintenance or controlled access purposes.

Experimental evaluation demonstrated reliable intrusion detection, stable cloud synchronisation, and effective operation of the alarm and monitoring

subsystems. The results indicate that Project VANTA provides a practical, low-cost, and scalable security solution suitable for residential, educational, commercial, and restricted-access environments.

Keywords: Internet of Things, ESP32, Laser Tripwire, Intrusion Detection, Cloud Monitoring, Security System, Blynk Platform, Remote Surveillance

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CHAPTER I

INTRODUCTION

1.1 Introduction

Security remains a fundamental requirement in residential, commercial, industrial, and institutional environments. The protection of personnel, assets, and restricted facilities has become increasingly important as modern infrastructures continue to expand in complexity and scale. Traditional security measures, including physical locks, passive barriers, and conventional alarm systems, often suffer from limitations associated with delayed response times, restricted accessibility, and high installation costs. Furthermore, many existing solutions do not provide continuous remote monitoring capabilities, thereby reducing their effectiveness in rapidly identifying and responding to security breaches.

Recent advancements in embedded systems and Internet of Things (IoT) technologies have enabled the development of intelligent monitoring systems capable of delivering real-time situational awareness. By combining sensor networks, wireless communication technologies, and cloud-based data services, IoT systems provide an effective means of monitoring environments remotely whilst maintaining relatively low implementation costs.

Project VANTA has been developed as an IoT-based laser tripwire security alarm system intended to provide reliable intrusion detection through optical sensing technology. The system establishes a virtual security perimeter by directing a continuous laser beam from a laser emitter module towards a laser receiver sensor. Under normal operating conditions, the receiver continuously detects the incoming beam, indicating that the protected zone remains secure. Any interruption of the optical path is interpreted as a potential intrusion event and immediately triggers the alarm subsystem.

The ESP32 microcontroller serves as the central processing unit of the system. In addition to monitoring the laser receiver, it manages alarm activation, display updates, cloud communication, and administrative control functions. When an intrusion is detected, the system activates an audible warning through an active piezoelectric buzzer, illuminates visual warning indicators, and updates a local liquid crystal display with the appropriate security status. Simultaneously,

system information is transmitted to a cloud-connected dashboard, enabling authorised users to observe system conditions remotely in real time.

A distinguishing feature of Project VANTA is the implementation of an administrative bypass mechanism. This functionality allows authorised personnel to temporarily suspend intrusion monitoring for a predetermined duration, facilitating maintenance activities and controlled access without triggering unnecessary alarms. Upon expiration of the bypass period, the system automatically restores normal monitoring operations.

The integration of embedded hardware, cloud communication services, and remote monitoring interfaces demonstrates the potential of IoT technologies within modern security applications. Project VANTA therefore serves as an example of how affordable hardware platforms can be utilised to develop intelligent, scalable, and user-friendly security solutions.

1.2 Objectives

The principal objectives of Project VANTA are as follows:

- To design and implement a laser-based intrusion detection system capable of identifying unauthorised access within a protected area.
- To develop a real-time monitoring solution using the ESP32 microcontroller and wireless communication technologies.
- To provide immediate audible and visual alerts whenever an intrusion event is detected.
- To display operational status information locally through a 16×2 I²C liquid crystal display.
- To facilitate remote monitoring through a cloud-connected dashboard accessible via the Internet.
- To implement an administrative bypass mechanism for authorised temporary suspension of the alarm system.
- To evaluate the effectiveness, reliability, and practicality of IoT-based security monitoring solutions.
- To demonstrate the integration of embedded systems, cloud computing, and wireless networking within a unified security platform.

1.3 Hardware and Software Used

Hardware Components

The principal hardware components utilised in Project VANTA are:

- ESP32 DevKit V1 Microcontroller Board
- Laser Emitter Module
- Laser Receiver Sensor Module
- Active Piezoelectric Buzzer
- 16×2 I²C Liquid Crystal Display
- Green Status LED
- Red Alert LEDs
- Breadboard Assembly
- Jumper Wires
- USB Power Supply

Software Components

The software technologies employed in the development of Project VANTA include:

- Arduino Integrated Development Environment (IDE)
- ESP32 Development Libraries
- Blynk Cloud Platform
- TypeScript
- Custom Web Dashboard
- Wi-Fi Communication Protocols

1.4 Block Diagram

The overall architecture of Project VANTA is illustrated in Figure 1.

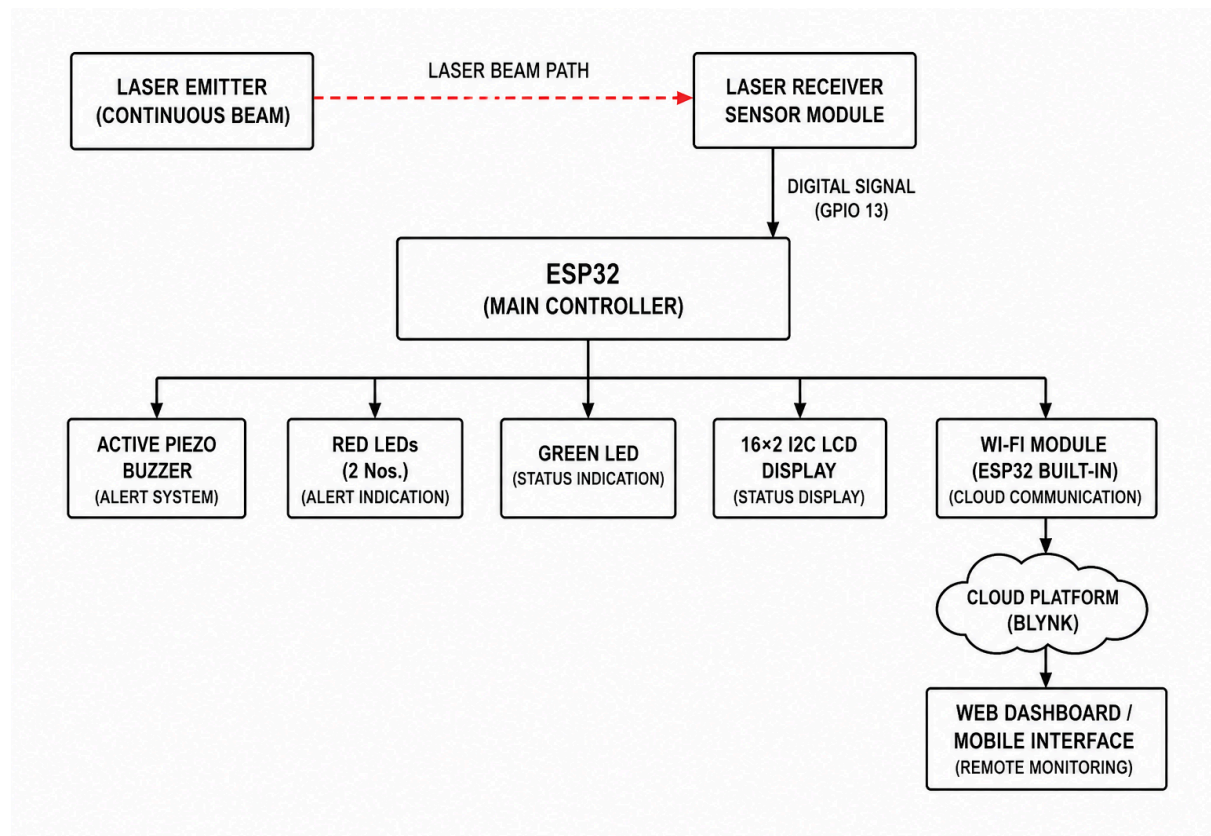


Figure 1: Block Diagram of Project VANTA

The system consists of four primary subsystems: the optical sensing subsystem, the processing subsystem, the alarm subsystem, and the cloud monitoring subsystem. The laser emitter continuously projects an optical beam towards the receiver sensor. The receiver status is monitored by the ESP32 microcontroller, which determines whether the security perimeter remains intact.

When an interruption of the optical path occurs, the ESP32 initiates a sequence of alarm responses including activation of the buzzer, illumination of visual indicators, and updating of the LCD display. Simultaneously, status information is transmitted through Wi-Fi to the Blynk cloud platform, where it is presented on the custom-developed monitoring dashboard. The dashboard additionally permits authorised users to activate the administrative bypass feature through secure cloud communication.

1.5 Chapter Summary

This chapter introduced Project VANTA and discussed the motivation behind the development of an IoT-enabled laser tripwire security system. The objectives, hardware and software requirements, and overall system architecture

were presented. The chapter established the foundation for the project by outlining the key components and functionalities of the proposed security platform. The subsequent chapter reviews existing literature and related technologies relevant to the development of Project VANTA.

CHAPTER II

LITERATURE SURVEY

The rapid advancement of Internet of Things (IoT) technologies has significantly influenced the development of intelligent security systems. Modern security solutions increasingly utilise interconnected sensors, embedded processors, cloud computing platforms, and wireless communication technologies to provide enhanced monitoring capabilities and real-time response mechanisms. This chapter reviews selected studies relevant to intrusion detection, IoT-based surveillance, and cloud-connected security systems.

2.1 IoT-Based Smart Security Systems

The application of IoT technologies within security systems has attracted considerable attention in recent years. IoT-enabled security platforms offer continuous monitoring, remote accessibility, and automated notification services, thereby overcoming many of the limitations associated with conventional security infrastructures.

Several researchers have proposed security architectures in which sensors continuously monitor environmental conditions and transmit information to cloud servers for analysis and visualisation. Such systems provide users with the ability to monitor protected areas remotely through web applications or mobile devices. The integration of wireless communication protocols further enhances system flexibility by eliminating the need for extensive physical infrastructure.

The principal advantage of IoT-based security systems lies in their ability to deliver real-time information whilst maintaining relatively low implementation costs. Consequently, these systems are increasingly adopted in residential properties, educational institutions, commercial premises, and industrial facilities.

2.2 Laser-Based Intrusion Detection Systems

Laser-based security systems represent an effective approach to perimeter monitoring. Such systems utilise a continuous optical beam to establish a virtual security boundary. Any interruption of the beam is immediately interpreted as a potential intrusion event, allowing the system to initiate alarm procedures.

Previous studies have demonstrated that laser-based detection systems offer several advantages over conventional motion sensors. These include rapid response times, reduced complexity, ease of installation, and improved detection accuracy within confined environments. Furthermore, laser tripwire systems can be deployed without substantial modifications to existing infrastructure.

However, traditional laser-based alarm systems are often limited by their lack of remote monitoring functionality. Many systems rely solely upon local alarm indicators, reducing their effectiveness when the monitored area is unattended. This limitation has motivated the integration of laser detection mechanisms with IoT communication technologies.

2.3 ESP32-Based Security Applications

The ESP32 microcontroller has emerged as one of the most widely utilised embedded platforms for IoT applications. Manufactured by Espressif Systems, the ESP32 combines processing capability, wireless communication functionality, and low power consumption within a compact and cost-effective development platform.

Numerous researchers have employed the ESP32 in smart security applications owing to its integrated Wi-Fi connectivity and versatile input/output interfaces. The device supports real-time sensor monitoring, cloud communication, local processing, and user interaction simultaneously. These characteristics make it particularly suitable for intrusion detection and surveillance applications.

Studies involving ESP32-based security systems have reported reliable performance, rapid processing speeds, and simplified integration with cloud services such as Blynk, MQTT platforms, and custom web applications.

2.4 Cloud-Based Monitoring and Remote Surveillance

Cloud computing has transformed the manner in which security systems are monitored and managed. Rather than relying exclusively upon local alarm mechanisms, modern systems can transmit operational data to cloud servers, enabling authorised users to access information remotely.

Cloud-connected dashboards provide several benefits, including real-time visualisation, historical event logging, remote control capabilities, and notification services. The Blynk platform has become particularly popular

within IoT development due to its simplicity, reliability, and support for multiple hardware platforms.

Remote monitoring significantly enhances the effectiveness of security systems by allowing users to observe system status regardless of their physical location. Consequently, cloud integration has become a standard feature within contemporary IoT security architectures.

2.5 Chapter Summary

The reviewed literature demonstrates that laser-based intrusion detection systems provide an effective method of identifying unauthorised access, whilst IoT technologies facilitate remote monitoring and real-time communication. The ESP32 platform offers an appropriate balance of processing capability, connectivity, and affordability for such applications.

Although previous studies have successfully implemented individual components of smart security systems, there remains a need for integrated solutions that combine optical intrusion detection, local alarm mechanisms, cloud-based monitoring, and administrative control functionality within a single platform. Project VANTA addresses this requirement through the development of a unified IoT-enabled laser tripwire security system.

CHAPTER III

IMPLEMENTATION

3.1 Introduction

The implementation phase of Project VANTA involved the integration of hardware components, embedded software, cloud communication technologies, and a web-based monitoring interface to develop a comprehensive IoT-enabled intrusion detection system. The project was designed to provide reliable real-time threat detection, local alarm generation, and remote monitoring capabilities whilst maintaining a simple and cost-effective architecture.

Project VANTA combines a laser-based optical detection mechanism with an ESP32 microcontroller, Blynk Cloud services, and a custom monitoring dashboard. The ESP32 serves as the central processing unit, continuously monitoring sensor inputs, controlling alarm outputs, updating the LCD display, and communicating with the cloud platform.

The implementation process consisted of hardware assembly, circuit development, firmware programming, cloud integration, and dashboard deployment. Each subsystem was developed and tested individually before being integrated into the complete security platform.

3.2 Hardware Components

3.2.1 ESP32 DevKit V1

The ESP32 DevKit V1 functions as the primary controller of Project VANTA. It is a dual-core microcontroller equipped with integrated Wi-Fi and Bluetooth capabilities, making it highly suitable for Internet of Things (IoT) applications.

The ESP32 is responsible for processing sensor inputs, executing intrusion detection algorithms, controlling alarm outputs, updating display information, and maintaining communication with the Blynk Cloud platform.



Figure 2: ESP32 DevKit V1

3.2.2 Laser Emitter Module

The laser emitter module generates a continuous optical beam that forms the virtual security boundary of the system. The emitter remains continuously active during system operation and directs the beam towards the laser receiver sensor.

Any interruption of this optical path is treated as a potential intrusion event.

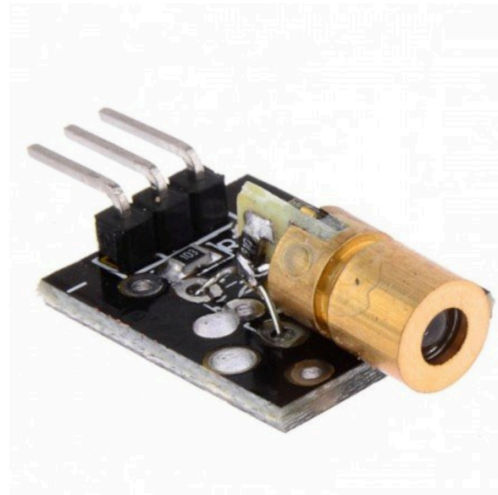


Figure 3: Laser Emitter Module

3.2.3 Laser Receiver Sensor Module

The laser receiver sensor continuously monitors the incoming laser beam and transmits its status to the ESP32 microcontroller.

When the beam remains uninterrupted, the protected area is considered secure. Any interruption causes a change in sensor output, triggering the intrusion detection process.

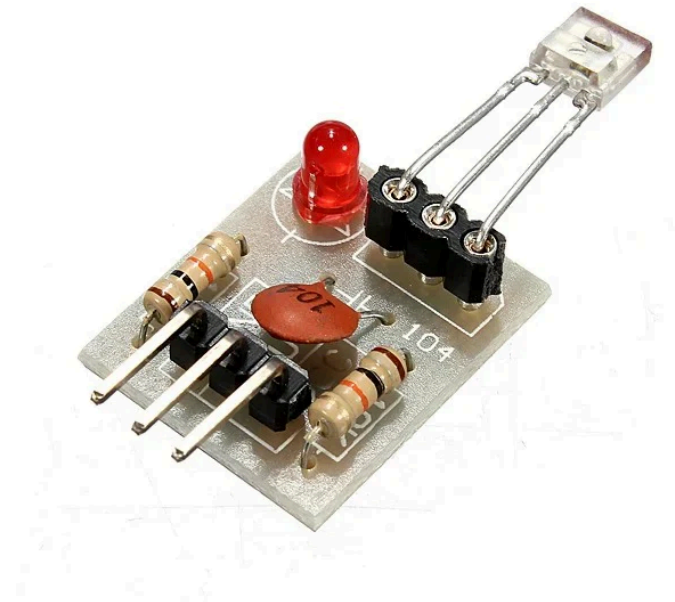


Figure 4: Laser Receiver Sensor Module

3.2.4 Active Piezoelectric Buzzer

The active piezoelectric buzzer serves as the primary audible alert mechanism of the system.

Upon detection of an intrusion event, the ESP32 activates the buzzer to generate an alarm signal intended to notify nearby individuals of a security breach.



Figure 5: Active Piezoelectric Buzzer

3.2.5 16×2 I²C Liquid Crystal Display

The LCD module provides visual feedback regarding the operational status of Project VANTA.

The display communicates with the ESP32 through the I²C communication protocol and presents information such as:

- System Initialisation
- Secure Status
- Intrusion Alerts
- Administrative Bypass Status
- Re-arm Countdown Information

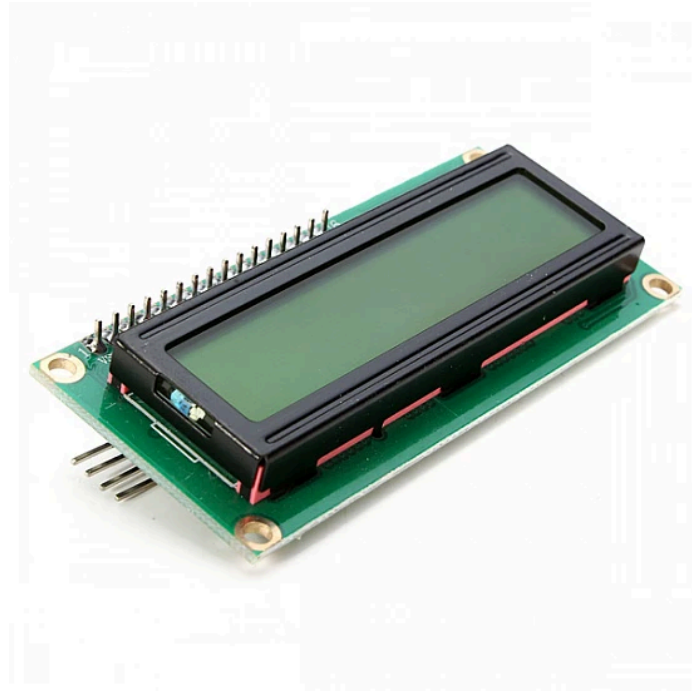


Figure 6: 16×2 I²C LCD Display

3.2.6 Status LEDs

Project VANTA incorporates one green LED and two red LEDs as visual indicators.

The green LED indicates normal secure operation, whilst the red LEDs flash alternately when an intrusion is detected. These indicators provide immediate visual confirmation of system status.

3.3 Software Components

3.3.1 Arduino Integrated Development Environment (IDE)

The Arduino Integrated Development Environment (IDE) was used for firmware development, compilation, debugging, and deployment to the ESP32 microcontroller.

The platform provides a user-friendly development environment and extensive support for ESP32-based applications.

3.3.2 ESP32 Development Libraries

A number of ESP32-specific libraries were utilised to support:

- Wi-Fi communication
- Cloud integration
- LCD interfacing
- GPIO management
- Timer operations
- These libraries simplified the implementation process and ensured reliable operation of the system.

3.3.3 Blynk Cloud Platform

The Blynk Cloud platform was utilised to provide remote monitoring and administrative control functionality for Project VANTA. Blynk is a widely used Internet of Things (IoT) platform that enables embedded devices to communicate with cloud services through wireless network connections. The platform facilitates real-time data exchange between the ESP32 microcontroller and authorised users, allowing system status information to be monitored remotely.

The ESP32 continuously transmits operational data to the Blynk Cloud server through its integrated Wi-Fi capability. This communication enables the system to synchronise intrusion events, security status updates, and administrative control commands in real time. As a result, users can monitor the condition of the protected area remotely without requiring direct physical access to the security system.

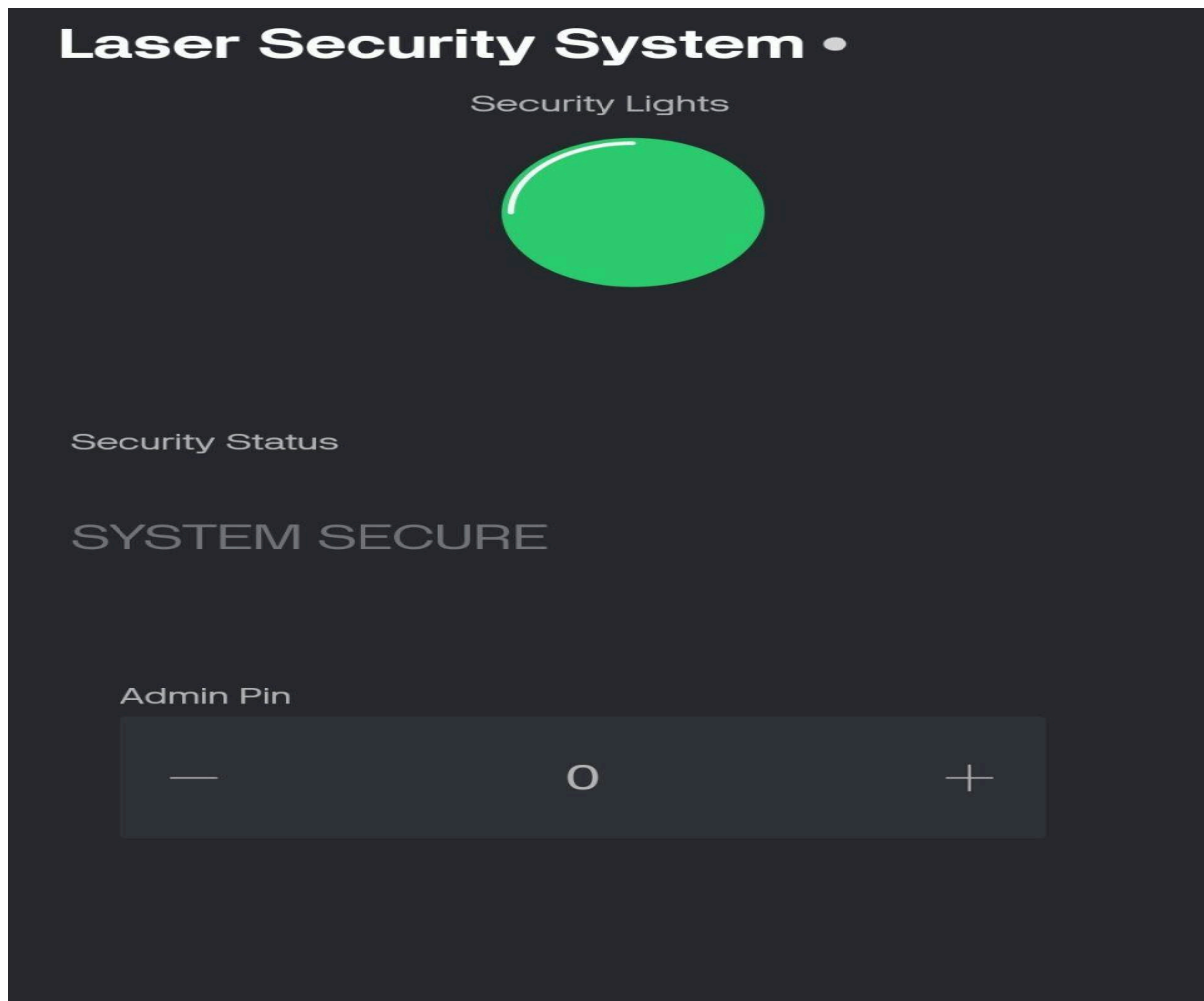


Figure 7: Blynk Mobile Monitoring Interface

The Blynk mobile application interface used during system testing is shown in Figure 7. The dashboard provides visual indication of the current security status through a graphical status indicator and displays system messages received from the ESP32. During normal operation, the interface indicates that the system is secure, whilst intrusion events are immediately reflected through cloud synchronisation. The dashboard additionally incorporates an administrative PIN control mechanism that allows authorised users to activate the temporary bypass mode for maintenance activities or controlled access.

The integration of Blynk provides several advantages, including remote accessibility, real-time monitoring, cloud-based communication, and simplified user interaction. The platform enables authorised personnel to supervise system activity from any location with internet connectivity whilst maintaining synchronisation with the local hardware components.

By combining the ESP32 microcontroller with the Blynk Cloud platform, Project VANTA achieves effective remote monitoring capabilities and extends the functionality of the security system beyond traditional local alarm mechanisms. The platform therefore serves as a critical component within the overall IoT architecture of the proposed intrusion detection system.

3.3.4 Custom Web Dashboard

A custom dashboard was developed to provide a modern and intuitive monitoring interface for Project VANTA.

The dashboard provides:

- Live system status updates
- Threat notifications
- Cloud connectivity information
- Administrative bypass controls
- Real-time telemetry monitoring



Figure 8: Project VANTA Dashboard Home Page

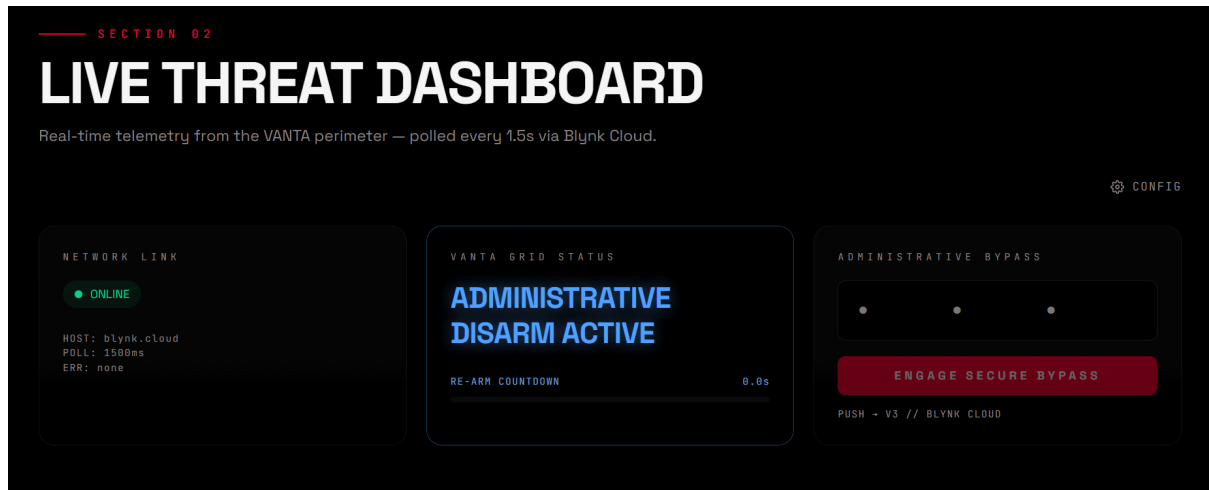


Figure 9: Project VANTA Dashboard Interface

Figure 9 illustrates the main monitoring interface of the custom web dashboard developed for Project VANTA. The dashboard was designed and implemented by the project team to provide a user-friendly platform for monitoring system activity and security status information in real time. Security data received from the ESP32 through cloud communication services is displayed within the interface, providing users with immediate visibility of system status, threat alerts, and operational information.

The dashboard complements the Blynk mobile application by providing an additional monitoring interface that enhances accessibility and user convenience. Through this interface, authorised users can remotely monitor system activity and review security status information from any internet-connected device.

3.4 Circuit Design and Hardware Connections

The hardware architecture of Project VANTA was designed to provide reliable operation whilst maintaining simplicity and ease of deployment.

The ESP32 acts as both the processing unit and the central power distribution hub. A single USB power source supplies the entire system, eliminating the need for external batteries.

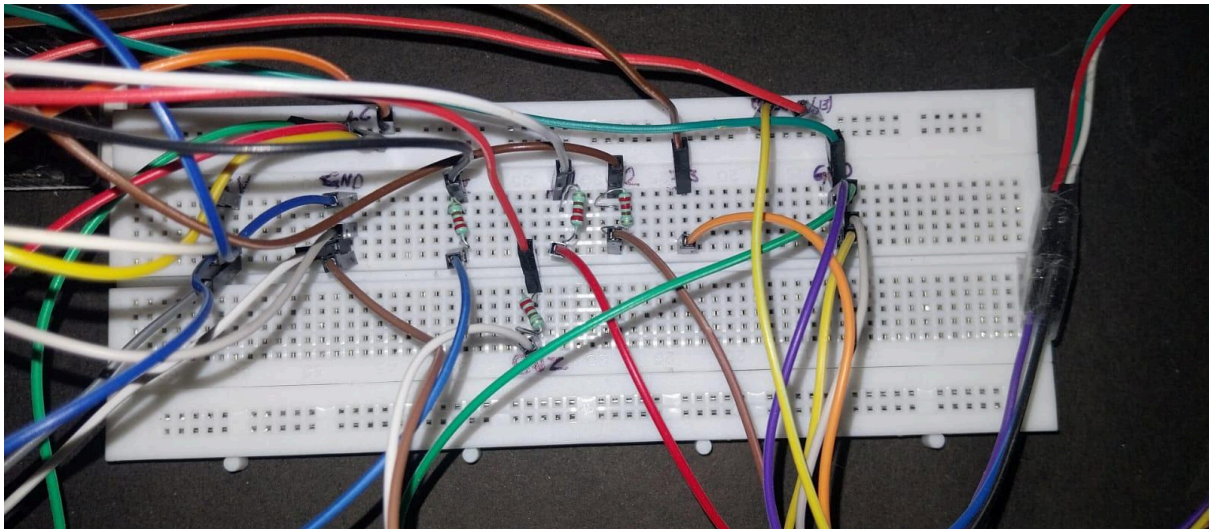


Figure 10: Prototype Wiring and Hardware Connections of Project VANTA

Table 3.1 Hardware Connections

Component	Connection
Laser Receiver Sensor	GPIO 13
Active Piezoelectric Buzzer	GPIO 12
Green Status LED	GPIO 14
Red Alert LED 1	GPIO 27
Red Alert LED 2	GPIO 26
LCD SDA	GPIO 21

LCD SCL

GPIO 22

The laser emitter operates continuously and remains aligned with the receiver sensor to establish the optical security perimeter.

A common ground architecture was adopted throughout the system to ensure stable electrical operation and minimise interference.

3.5 Firmware Development

The firmware developed for Project VANTA serves as the operational core of the system. It is responsible for managing all system functions, including sensor monitoring, intrusion detection, alarm activation, cloud communication, LCD updates, dashboard synchronisation, and administrative bypass management.

The software was developed using the Arduino programming framework and deployed to the ESP32 microcontroller.

The firmware continuously monitors the laser receiver sensor and analyses its output state. Under secure operating conditions, the system maintains normal monitoring mode and updates the dashboard accordingly.

When interruption of the laser beam is detected, the ESP32 immediately initiates the alarm sequence. This includes activating the buzzer, flashing the alert LEDs, updating the LCD display, and transmitting an intrusion alert to the cloud platform.

A simplified representation of the main execution loop is shown below:

```
void loop()
{
    Blynk.run();
    timer.run();
}
```

The firmware additionally incorporates an administrative bypass mechanism that allows authorised users to temporarily disable intrusion monitoring through the dashboard. Following expiration of the bypass period, the system automatically re-arms itself and resumes normal operation.

The software architecture was designed to ensure reliable real-time operation whilst maintaining stable communication with the cloud infrastructure.

3.6 Complete Program Listing

The complete firmware developed for Project VANTA is presented in this section. The source code implements system initialisation, Wi-Fi communication, Blynk Cloud integration, intrusion detection, alarm control, LCD management, dashboard synchronisation, and administrative bypass functionality.

ESP32 Firmware Source Code

```
#define BLYNK_TEMPLATE_ID "TMPL3b7k1FGY9"

#define BLYNK_TEMPLATE_NAME "Laser Security System"

#define BLYNK_AUTH_TOKEN "2cUgXrf_H7zaWRxCLC7-6R1IFBtbhuQ7"


#include <WiFi.h>

#include <WiFiClient.h>

#include <BlynkSimpleEsp32.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>


char ssid[] = "YOUR_WIFI_NAME";

char pass[] = "YOUR_WIFI_PASSWORD";
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);

const int LASER_RECEIVER = 13;

const int BUZZER      = 12;

const int GREEN_LED   = 14;

const int RED_LED_1   = 27;

const int RED_LED_2   = 26;


const int SECRET_PIN  = 1234;

const unsigned long BYPASS_TIME = 30; // 30 Seconds Admin Disarm Window


bool isBypassed = false;

unsigned long bypassStartTime = 0;

int lastRemainingSeconds = -1;


BlynkTimer timer;

unsigned long previousMillis = 0;

const long blinkInterval = 500;

bool dynamicBlinkState = false;

String lastCloudStatus = "";


BLYNK_WRITE(V3) {

    int enteredPin = param.asInt();

    if (isBypassed) return;
```

```
if (enteredPin == SECRET_PIN) {  
    isBypassed = true;  
    bypassStartTime = millis();  
    lastRemainingSeconds = -1;  
    Serial.println("[EVENT] Correct PIN entered. Initializing countdown tracks...");  
}  
}  
  
void checkSecuritySystem() {  
    unsigned long currentMillis = millis();  
    if (currentMillis - previousMillis >= blinkInterval) {  
        previousMillis = currentMillis;  
        dynamicBlinkState = !dynamicBlinkState;  
    }  
  
    if (isBypassed) {  
        unsigned long elapsedSeconds = (millis() - bypassStartTime) / 1000;  
        if (elapsedSeconds < BYPASS_TIME) {  
            int remainingSeconds = BYPASS_TIME - elapsedSeconds;  
            digitalWrite(BUZZER, LOW);  
            digitalWrite(RED_LED_1, LOW);  
            digitalWrite(RED_LED_2, LOW);  
            digitalWrite(GREEN_LED, HIGH);  
        }  
    }  
}
```

```
if (lastCloudStatus != "BYPASS") {  
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print("ADMIN BYPASS ON ");  
    Blynk.virtualWrite(V1, "ADMIN BYPASS ACTIVE");  
    Blynk.setProperty(V0, "color", "#3498DB");  
    Blynk.virtualWrite(V0, 255);  
    lastCloudStatus = "BYPASS";  
}  
  
if (remainingSeconds != lastRemainingSeconds) {  
    lastRemainingSeconds = remainingSeconds;  
    int mins = remainingSeconds / 60;  
    int secs = remainingSeconds % 60;  
    lcd.setCursor(0, 1);  
    char timeBuffer[32];  
    sprintf(timeBuffer, "RE-ARM IN: %02d:%02d", mins, secs);  
    lcd.print(timeBuffer);  
}  
return;  
} else {  
    isBypassed = false;  
    lastCloudStatus = "";  
    lcd.clear();
```

```
Blynk.virtualWrite(V3, 0);

Serial.println("[EVENT] Countdown complete. Re-arming threat detection rails.");

}

}

int receiverState = digitalRead(LASER_RECEIVER);

if (receiverState == HIGH) {

    digitalWrite(GREEN_LED, LOW);

    digitalWrite(BUZZER, HIGH);

    digitalWrite(RED_LED_1, dynamicBlinkState);

    digitalWrite(RED_LED_2, !dynamicBlinkState);

    if (lastCloudStatus != "ALERT") {

        lcd.clear();

        lcd.setCursor(0, 0);

        lcd.print("  !! ALERT !! ");

        lcd.setCursor(0, 1);

        lcd.print("TRIPWIRE BROKEN ");

        Blynk.virtualWrite(V1, "BREACH DETECTED!");

        Blynk.setProperty(V0, "color", "#D32F2F");

        Blynk.virtualWrite(V0, 255);

        lastCloudStatus = "ALERT";

    }

} else {
```

```
digitalWrite(GREEN_LED, HIGH);

digitalWrite(BUZZER, LOW);

digitalWrite(RED_LED_1, LOW);
digitalWrite(RED_LED_2, LOW);


if (lastCloudStatus != "SECURE") {

    lcd.clear();

    lcd.setCursor(0, 0);

    lcd.print("SYSTEM STATUS: ");

    lcd.setCursor(0, 1);

    lcd.print("  SECURE  ");

    Blynk.virtualWrite(V1, "SYSTEM SECURE");

    Blynk.setProperty(V0, "color", "#2ECC71");

    Blynk.virtualWrite(V0, 255);

    lastCloudStatus = "SECURE";

}

}

}

void setup() {

    Serial.begin(115200);

    pinMode(LASER_RECEIVER, INPUT);

    pinMode(BUZZER, OUTPUT);

    pinMode(GREEN_LED, OUTPUT);
```

```
pinMode(RED_LED_1, OUTPUT);  
  
pinMode(RED_LED_2, OUTPUT);  
  
  
lcd.init();  
  
lcd.backlight();  
  
lcd.setCursor(0, 0);  
  
lcd.print("PROJECT VANTA ");  
  
lcd.setCursor(0, 1);  
  
lcd.print("INITIALIZING...");  
  
  
  
Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);  
  
lcd.clear();  
  
timer.setInterval(100L, checkSecuritySystem);  
}  
  
void loop() {  
  
    Blynk.run();  
  
    timer.run();  
  
}
```

The firmware constitutes the software foundation of Project VANTA and coordinates all interactions between hardware devices, cloud services, and user interfaces.

3.7 System Architecture

Project VANTA follows a layered architecture comprising four principal subsystems:

- Optical Detection Subsystem
- Embedded Processing Subsystem
- Alarm and Notification Subsystem
- Cloud Monitoring Subsystem

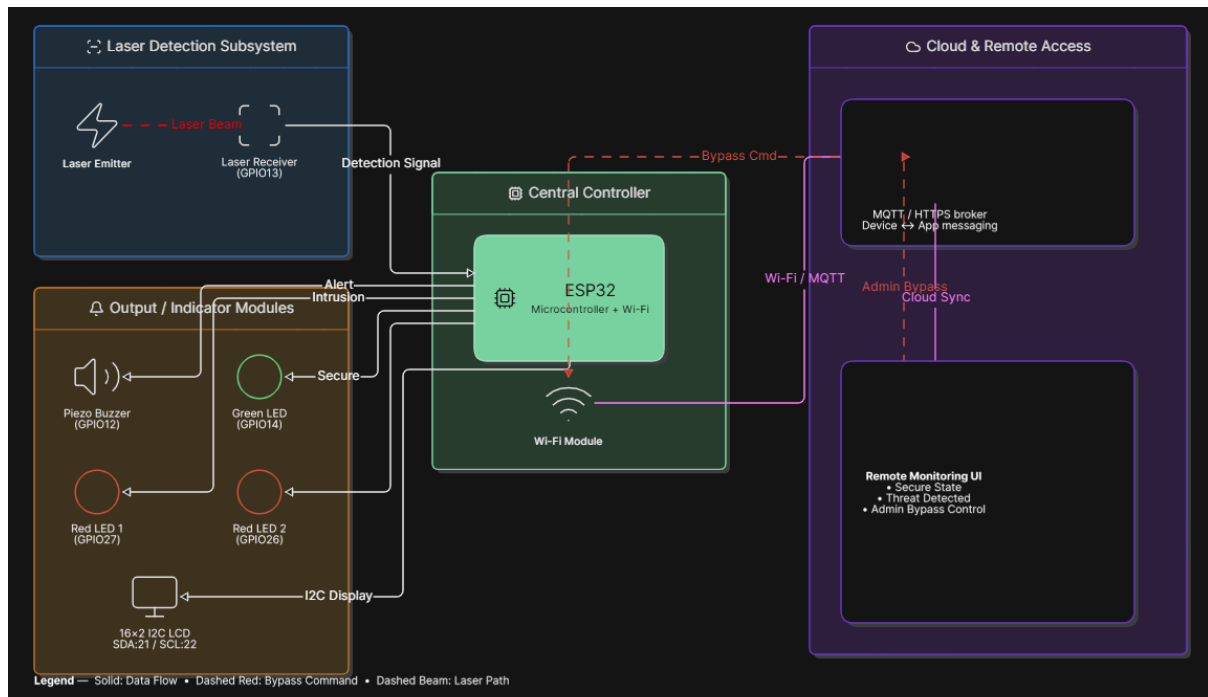


Figure 11: Project VANTA System Architecture

The Optical Detection Subsystem establishes and monitors the laser boundary. The Embedded Processing Subsystem analyses sensor data and executes the intrusion detection algorithm.

The Alarm and Notification Subsystem provides immediate local feedback through the buzzer, LEDs, and LCD display. The Cloud Monitoring Subsystem facilitates remote supervision and administrative control through the Blynk platform and custom dashboard.

The interaction between these subsystems, as illustrated in Figure 11, enables Project VANTA to operate as a unified IoT-based security solution capable of providing both local and remote threat monitoring capabilities.

3.8 Chapter Summary

This chapter presented the implementation details of Project VANTA, including hardware selection, software tools, circuit design, firmware development, cloud

integration, and overall system architecture. The successful integration of embedded hardware, wireless networking technologies, cloud communication services, and web-based monitoring capabilities resulted in a functional and reliable intrusion detection platform.

The subsequent chapter discusses the working methodology, system testing procedures, obtained results, and performance analysis of the developed security system.

CHAPTER IV

WORKING METHODOLOGY, RESULTS AND ANALYSIS

4.1 Introduction

This chapter presents the operational methodology, testing procedures, experimental results, and performance analysis of Project VANTA. Following the implementation phase described in the previous chapter, the completed system was subjected to a series of functional tests to verify the reliability of intrusion detection, alarm activation, cloud communication, dashboard synchronisation, and administrative bypass functionality.

The objective of testing was to ensure that all hardware and software components operated together as an integrated security platform capable of detecting intrusion events and providing real-time monitoring capabilities.

4.2 Working Methodology

Project VANTA operates by establishing a continuous optical security perimeter using a laser emitter and laser receiver sensor. During normal operation, the laser beam remains aligned with the receiver, indicating that the protected area is secure.

The ESP32 continuously monitors the output of the laser receiver sensor. When the optical path is interrupted, the sensor output changes state and an intrusion event is immediately detected.

Upon detection of a breach, the ESP32 performs the following actions:

1. Activates the piezoelectric buzzer.
2. Initiates alternating flashing of the red alert LEDs.
3. Updates the LCD display with an intrusion warning message.
4. Transmits an alert notification to the Blynk Cloud platform.
5. Updates the web dashboard to indicate an active security breach.

When the optical path is restored, the system automatically returns to secure monitoring mode and updates all connected interfaces accordingly.

The system additionally incorporates an administrative bypass feature that temporarily disables intrusion monitoring when a valid access code is entered through the dashboard interface.

4.3 Intrusion Detection Process

The intrusion detection process begins with continuous monitoring of the laser receiver sensor.

Under secure conditions, the laser beam remains uninterrupted and the system maintains its normal monitoring state. The green status LED remains illuminated, the buzzer remains inactive, and the LCD display indicates secure operation.

When the beam is interrupted, the receiver immediately signals the ESP32. The firmware processes this event and activates the alarm subsystem.

The response sequence includes the following actions:

1. Generation of an audible alarm through the piezoelectric buzzer.
2. Activation of visual alert indicators.
3. Display of a warning message on the LCD module.
4. Transmission of an alert notification to the Blynk Cloud platform.
5. Updating of the web dashboard to reflect the intrusion event.

This event-driven architecture enables Project VANTA to respond rapidly to security breaches whilst maintaining continuous monitoring capabilities.

4.4 Cloud Communication and Dashboard Integration

The cloud communication subsystem was implemented using the Blynk Cloud platform.

The ESP32 establishes a secure Wi-Fi connection and exchanges information with the cloud server through encrypted communication channels. System status information is transmitted continuously, allowing authorised users to monitor the security platform remotely.

The dashboard receives and displays:

- System operational status
- Intrusion alerts

- Administrative bypass status
- Cloud connectivity information
- Telemetry updates

The average dashboard refresh interval was configured to approximately 1.5 seconds, ensuring near real-time visibility of system events.

This integration significantly enhances the functionality of Project VANTA by extending monitoring capabilities beyond the physical installation site.

4.5 Administrative Bypass Operation

An administrative bypass mechanism was implemented to permit authorised temporary access to the protected area without generating intrusion alarms.

When the correct security code is entered through the dashboard interface, the ESP32 activates bypass mode and suspends intrusion detection for a predefined period.

During bypass operation:

- Alarm outputs remain disabled.
- The LCD display presents bypass information.
- Dashboard status is updated accordingly.
- A countdown timer indicates the remaining bypass duration.

Upon expiration of the bypass interval, the system automatically re-arms itself and resumes normal intrusion monitoring.

This functionality improves usability whilst maintaining overall system security.

4.6 Dashboard Implementation and Monitoring Interface

A dedicated web-based monitoring interface was developed to provide real-time visualisation of Project VANTA system activity.

The dashboard communicates with the ESP32 through the Blynk Cloud platform and displays operational information in a structured and user-friendly format.

Dashboard Features

The dashboard provides the following functionality:

- Live system status monitoring
- Cloud connectivity verification
- Telemetry information display
- Administrative bypass control
- Real-time threat visualisation

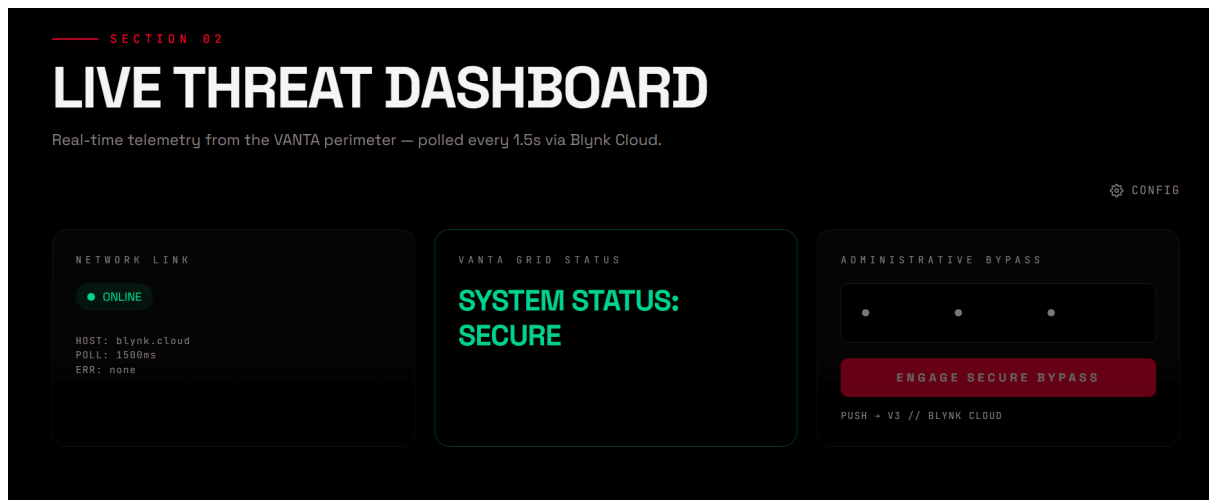


Figure 12: Project VANTA Dashboard – Secure State

Figure 12 illustrates the dashboard during normal system operation. The interface displays a secure status, confirms active cloud connectivity, and provides real-time telemetry updates from the ESP32. This view allows users to verify that the protected area remains secure and that all monitoring functions are operating correctly.

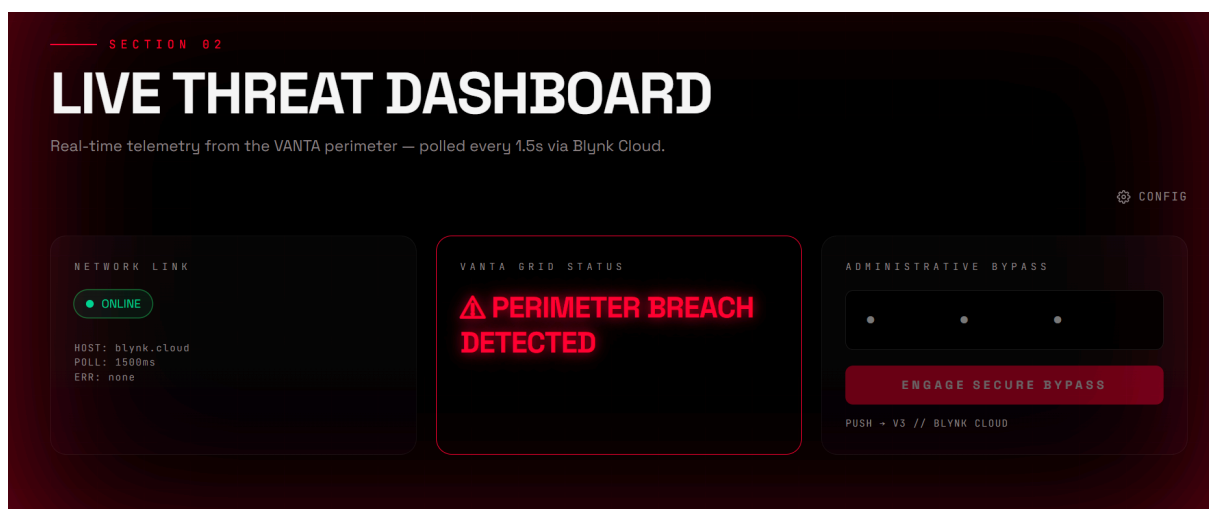


Figure 13: Project VANTA Dashboard – Threat Detection State

Figure 13 demonstrates the dashboard response when the laser security perimeter is interrupted. Upon intrusion detection, the dashboard immediately updates the system status and provides visual indication of the detected threat, allowing rapid identification of security breaches.

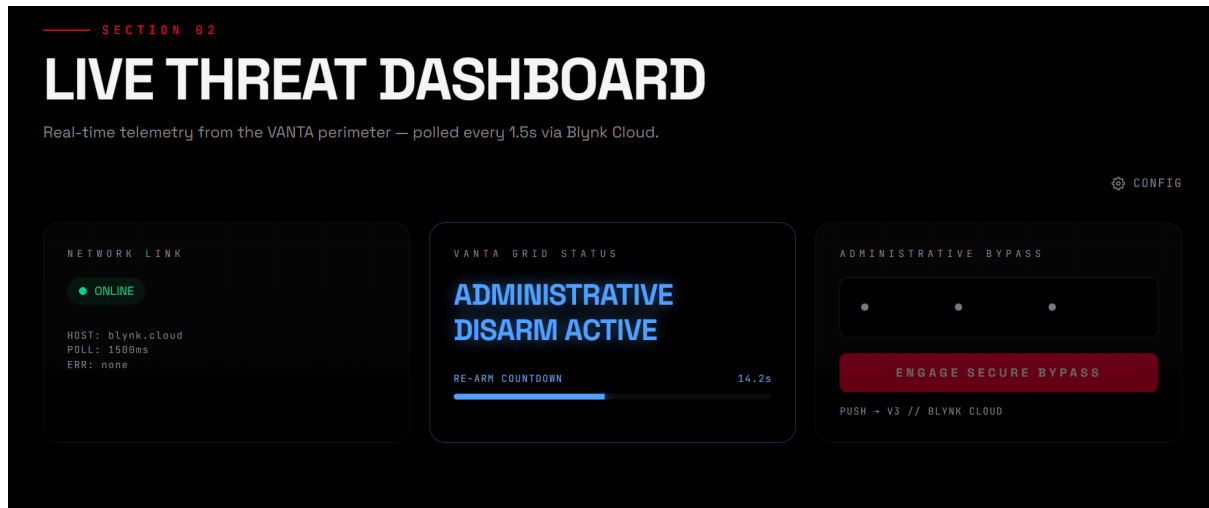
*Figure 14: Project VANTA Dashboard – Administrative Bypass Mode*

Figure 14 shows the administrative bypass feature, which enables authorised users to temporarily disable intrusion monitoring for maintenance or authorised access purposes. This functionality provides operational flexibility whilst maintaining cloud communication and dashboard accessibility.

The implementation of this dashboard enhances situational awareness and enables remote supervision of the security platform from any internet-connected device.

4.7 Experimental Testing

A series of experimental tests were conducted to evaluate the performance and reliability of Project VANTA.

The testing process focused on:

- Intrusion detection accuracy
- Alarm activation
- Cloud communication
- Dashboard synchronisation

- Administrative bypass functionality

Table 4.1 Experimental Test Cases

Test Case	Expected Result	Outcome
System Power-On	System Initialises Successfully	Successful
Laser Beam Aligned	Secure Status Displayed	Successful
Laser Beam Interrupted	Alarm Activated	Successful
Dashboard Synchronisation	Real-Time Status Update	Successful
Administrative Bypass Activation	Monitoring Temporarily Disabled	Successful
Automatic Re-Arming	Monitoring Restored	Successful

The results indicate that all primary system functions operated as intended during testing.

4.8 Results and Performance Analysis

The testing results demonstrate that Project VANTA successfully achieved its design objectives.

The intrusion detection mechanism reliably identified interruptions in the laser beam and generated immediate local and remote responses. Alarm activation occurred consistently during all intrusion tests, whilst dashboard updates were reflected within a short period following detection.

The Blynk Cloud integration provided stable communication throughout testing, enabling effective remote monitoring and control. No significant communication failures were observed under normal operating conditions.

The administrative bypass feature functioned correctly and automatically restored monitoring following expiration of the configured bypass interval.

Overall, the system exhibited reliable performance across all tested operating scenarios.

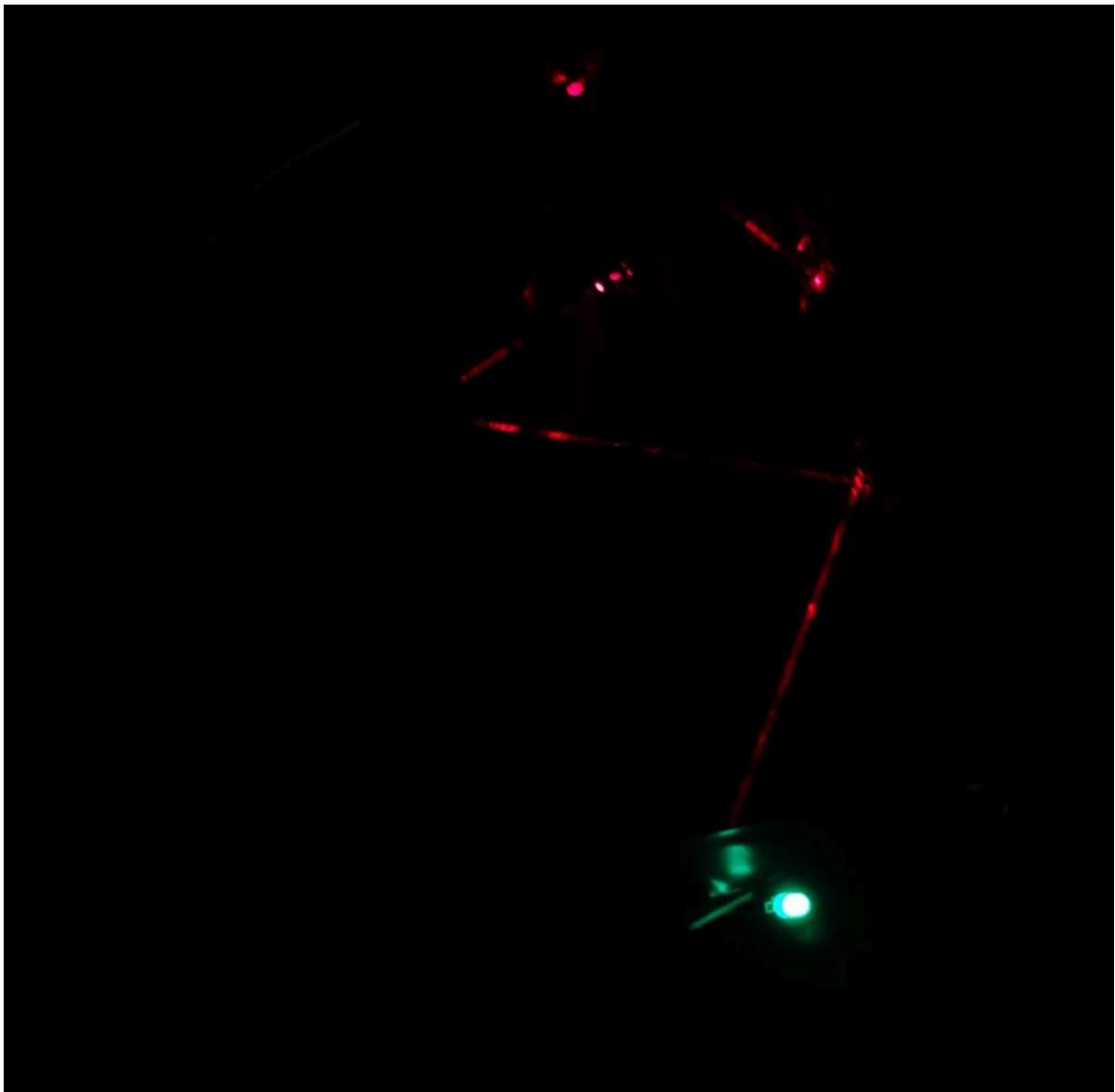


Figure 15: Project VANTA Prototype During Secure Operation (Green LED ON)

Figure 15 shows the prototype operating under normal conditions. The green status indicator confirms that the laser beam remains uninterrupted and the protected area is secure. During this state, the system continuously monitors the environment whilst maintaining communication with the Blynk Cloud platform.



Figure 16: Project VANTA Prototype During Threat Detection (Red Alert LEDs ON)

Figure 16 illustrates the system response following intrusion detection. When the laser beam is interrupted, the red alert indicators are activated and the alarm mechanism is triggered. Simultaneously, the ESP32 transmits the event to the cloud platform, allowing the dashboard to display the updated threat status in near real-time.

Table 4.2 Performance Analysis

Parameter	Observed Performance
Intrusion Detection	Reliable
Alarm Response	Immediate
Dashboard Synchronisation	Near Real-Time
Cloud Connectivity	Stable
Administrative Bypass	Functional
System Reliability	High

The experimental results confirm that Project VANTA is capable of functioning as an effective IoT-enabled intrusion detection and monitoring platform. The combination of local alarm activation, cloud connectivity, remote monitoring capabilities, and administrative control features demonstrates the suitability of the system for small-scale security applications and educational IoT deployments.

4.9 Advantages of Project VANTA

The developed system offers several advantages:

- Low-cost implementation
- Real-time intrusion detection
- Remote monitoring capabilities
- Cloud-based communication
- User-friendly dashboard interface
- Administrative bypass functionality
- Simple installation and deployment
- Scalable architecture for future expansion

These characteristics make the system suitable for residential, educational, commercial, and small-scale industrial applications.

4.10 Limitations of the System

Despite its successful operation, several limitations were identified.

The current implementation utilises a single laser detection zone and therefore protects only one perimeter line at a time. Additionally, the system depends upon a stable Wi-Fi connection for cloud communication and remote monitoring.

Environmental factors such as sensor misalignment or excessive ambient light may also influence system performance under certain conditions.

These limitations provide opportunities for future enhancement and development.

4.11 Chapter Summary

This chapter presented the working methodology, testing procedures, experimental results, and performance analysis of Project VANTA. The obtained results demonstrate that the system successfully performs intrusion detection, alarm activation, cloud communication, dashboard synchronisation, and administrative bypass operations. The overall performance of the developed platform confirms its suitability as a practical and cost-effective IoT-based security solution.

The following chapter presents the conclusions drawn from the project and discusses potential future enhancements to the system.

CHAPTER V

CONCLUSION AND FUTURE WORK

5.1 Conclusion

The objective of this project was to design, develop, and evaluate an Internet of Things (IoT)-enabled laser tripwire security alarm system capable of detecting unauthorised intrusion and providing real-time monitoring through cloud connectivity. The resulting system, Project VANTA, successfully integrates optical sensing technology, embedded processing, wireless communication, and remote monitoring within a unified security platform.

The developed system utilises a laser emitter and receiver arrangement to establish a virtual security perimeter. Continuous monitoring of the optical link is performed by the ESP32 microcontroller, which acts as the central processing unit responsible for intrusion detection, alarm management, cloud communication, and user interaction. Upon interruption of the laser beam, the system immediately initiates both audible and visual alarm responses whilst simultaneously transmitting status information to a cloud-based monitoring dashboard.

Experimental testing demonstrated that the system was capable of reliably detecting intrusion events and generating appropriate local and remote notifications. The integration of the Blynk Cloud platform enabled real-time synchronisation between the physical security system and the web-based monitoring interface, thereby extending the accessibility and functionality of the solution beyond that of conventional standalone alarm systems.

Furthermore, the implementation of an administrative bypass mechanism enhanced system usability by permitting authorised temporary access without compromising overall security. The countdown-based reactivation process ensured that the system automatically returned to its protected state following authorised use.

From an engineering perspective, Project VANTA successfully demonstrates the practical application of embedded systems, wireless networking, cloud computing, and sensor technologies within a modern security environment. The project confirms that effective security monitoring solutions can be developed

using affordable hardware whilst maintaining reliability, scalability, and ease of deployment.

Overall, the objectives established at the commencement of the project were achieved successfully. The system provides a robust proof-of-concept implementation of an IoT-enabled intrusion detection platform and highlights the potential of low-cost smart security technologies for residential, educational, commercial, and industrial applications.

5.2 Future Work

Whilst Project VANTA achieved its primary objectives, several opportunities exist for further enhancement and expansion.

One potential improvement would be the integration of camera-based surveillance capabilities. By incorporating an ESP32-CAM module or network-connected camera, the system could capture photographic evidence whenever an intrusion event is detected. Such functionality would significantly enhance incident verification and post-event analysis.

The implementation of automated notification services represents another valuable enhancement. Future versions of the system could transmit alerts through Short Message Service (SMS), electronic mail, or mobile push notifications, thereby ensuring that authorised users receive immediate warnings regardless of whether the dashboard is actively being monitored.

Artificial Intelligence (AI) and machine learning technologies may also be incorporated to improve threat assessment and reduce false alarms. Intelligent algorithms could analyse intrusion patterns and distinguish between genuine security threats and non-critical environmental disturbances.

The current implementation utilises a single laser boundary. Future developments could support multiple laser zones, enabling larger areas to be protected simultaneously. Such an approach would be particularly beneficial for warehouses, laboratories, industrial facilities, and restricted-access environments.

Another area for future investigation involves the incorporation of backup power systems. The addition of rechargeable battery support would allow the

system to remain operational during power interruptions, thereby improving reliability in critical security applications.

Further development could also focus on integration with smart building and home automation platforms. Compatibility with existing automation ecosystems would permit coordinated responses between security devices, lighting systems, access control mechanisms, and surveillance infrastructure.

Finally, future iterations may incorporate secure authentication mechanisms such as biometric verification, RFID access control, or facial recognition technologies to further strengthen access management and administrative security.

The proposed enhancements would significantly extend the capabilities of Project VANTA and contribute towards the development of a more comprehensive intelligent security ecosystem.

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APPENDIX A

THREAT ALERT TESTING PHOTOGRAPHS

This appendix presents photographic evidence of the Threat Alert module during system testing. The images demonstrate the operation of the laser tripwire detection system and the corresponding alert mechanism when an intrusion event is detected.

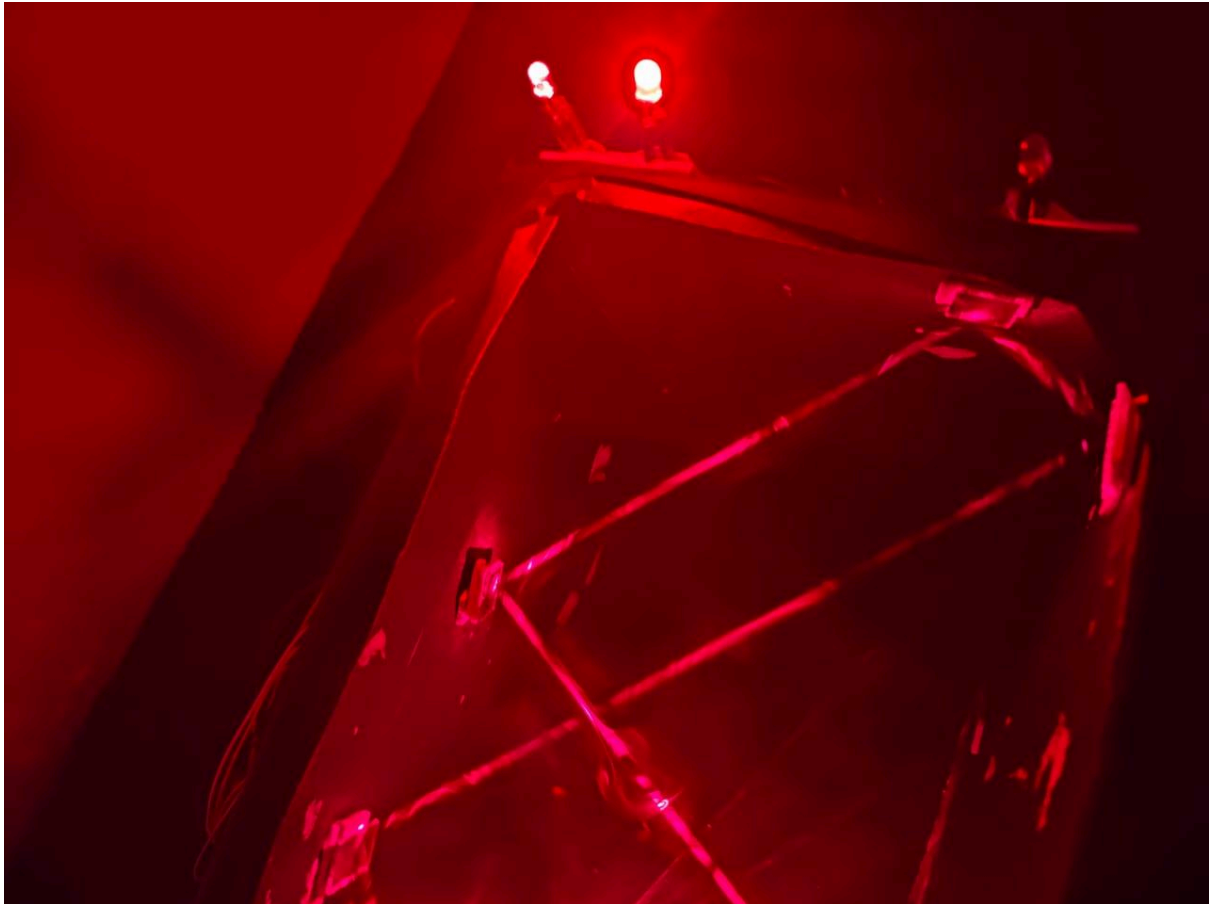


Figure A.1 – Laser Tripwire Detection System During Testing

This photograph shows the laser-based tripwire detection setup operating under test conditions. Multiple laser beams are projected across the protected area to establish an intrusion detection zone. The visible laser beams indicate that the monitoring system is active and continuously monitoring for interruptions.



Figure A.2 – Threat Alert Notification Following Tripwire Interruption

This photograph shows the alert generated by the system following the interruption of a laser tripwire. The LCD display presents the message “!! ALERT !! TRIPWIRE BROKEN”, confirming that the interruption was successfully detected and processed by the system. This demonstrates the correct operation of the threat detection and alert notification functionality.